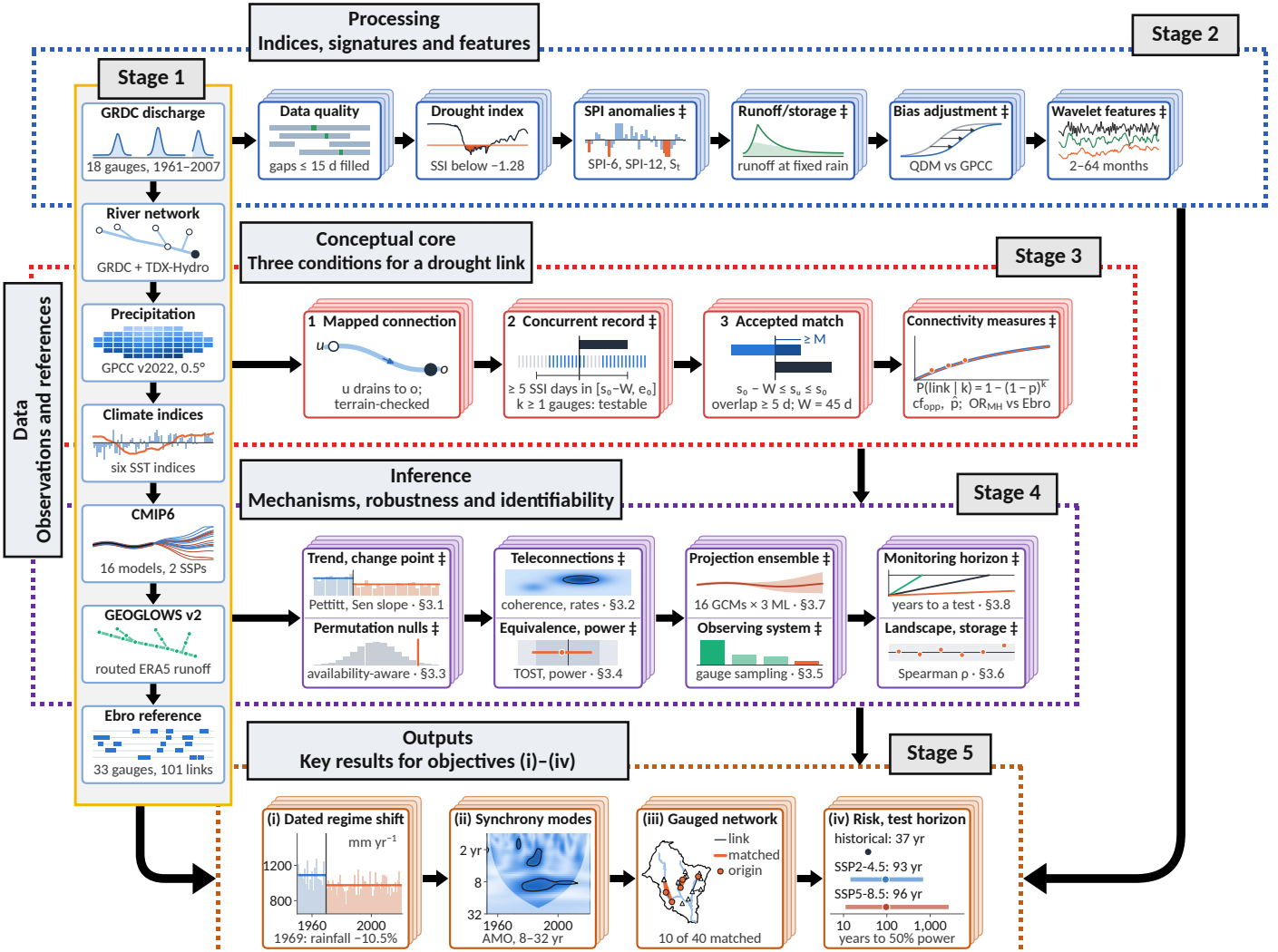

Data
Observations and references
Stage 1
GRDC discharge
18 gauges, 1961–2007
River network
GRDC + TDX-Hydro
Precipitation
GPCC v2022, 0.5°
Climate indices
six SST indices
CMIP6
16 models, 2 SSPs
GEOGLOWS v2
routed ERA5 runoff
Ebro reference
33 gauges, 101 links
Processing
Indices, signatures and features
Stage 2
Data quality
gaps ≤ 15 d filled
Drought index
SSI below −1.28
SPI anomalies ‡
SPI-6, SPI-12, S_t
Runoff/storage ‡
runoff at fixed rain
Bias adjustment ‡
QDM vs GPCC
Wavelet features ‡
2–64 months
Conceptual core
Three conditions for a drought link
Stage 3
1 Mapped connection
u drains to o;
terrain-checked
2 Concurrent record ‡
≥ 5 SSI days in $[s_0-W, e_0]$
k ≥ 1 gauges: testable
3 Accepted match
≥ M
$s_0 - W \le s_u \le s_0$
overlap ≥ 5 d; W = 45 d
Connectivity measures ‡
$P(\text{link} \mid k) = 1-(1-p)^k$
cf_{opp}, $\hat{p}$; OR_{MH} vs Ebro
Inference
Mechanisms, robustness and identifiability
Stage 4
Trend, change point ‡
Pettitt, Sen slope · §3.1
Permutation nulls ‡
availability-aware · §3.3
Teleconnections ‡
coherence, rates · §3.2
Equivalence, power ‡
TOST, power · §3.4
Projection ensemble ‡
16 GCMs × 3 ML · §3.7
Observing system ‡
gauge sampling · §3.5
Monitoring horizon ‡
years to a test · §3.8
Landscape, storage ‡
Spearman ρ · §3.6
Outputs
Key results for objectives (i)–(iv)
Stage 5
(i) Dated regime shift
mm yr⁻¹
1200
800
1960
2000
1969: rainfall −10.5%
(ii) Synchrony modes
2 yr
8
32
1960
2000
AMO, 8–32 yr
(iii) Gauged network
link
matched
origin
10 of 40 matched
(iv) Risk, test horizon
historical: 37 yr
SSP2-4.5: 93 yr
SSP5-8.5: 96 yr
10
100
1,000
years to 50% power